

El problema es

$$\begin{aligned}u_t - \kappa u_{xx} &= 0, & 0 < x < L, \\u(0, t) &= u(L, t) = 0, \\u(x, 0) &= T_0 \sin\left(\frac{\pi x}{L}\right).\end{aligned}$$

Separamos

$$u(x, t) = X(x)T(t).$$

Obtenemos

$$\frac{T'}{T} = \kappa \frac{X''}{X},$$

es decir

$$\begin{aligned}T' + \lambda T &= 0, \\X'' + k^2 X &= 0,\end{aligned}$$

con

$$k^2 = \frac{\lambda}{\kappa}.$$

Luego

$$\begin{aligned}T(t) &= e^{-\lambda t}, \\X(x) &= A \cos(kx) + B \sin(kx).\end{aligned}$$

Imponemos CC:

$$\begin{aligned}X(0) &= A = 0, \\X(L) &= B \sin(kL) = 0,\end{aligned}$$

luego

$$\begin{aligned}k &= k_n = \frac{n\pi}{L}, \\\lambda &= \lambda_n = \kappa k_n^2 = \kappa \left(\frac{n\pi}{L}\right)^2\end{aligned}$$

y

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin(k_n x) e^{-\lambda_n t}.$$

Imponemos CI:

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin(k_n x) \sum_{n=1}^{\infty} B_n \sin\left(n \frac{\pi x}{L}\right) = T_0 \sin\left(\frac{\pi x}{L}\right),$$

luego por ortogonalidad y completitud de los senos y cosenos en $[0, L]$ queda

$$B_1 = T_0,$$

$$B_n = 0, \quad n = 2, 3, \dots$$

En definitiva

$$u(x, t) = T_0 \sin\left(\frac{\pi x}{L}\right) \exp\left(-\kappa \frac{\pi^2}{L^2} t\right).$$